Introduction to Linear Programming

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Learning Objectives

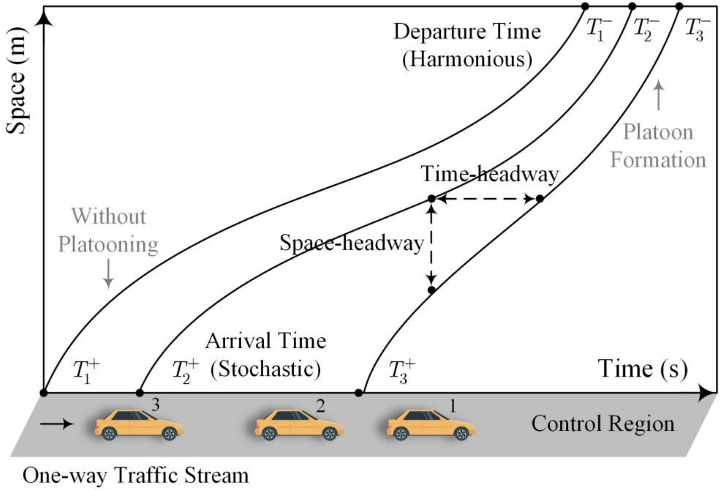
- Basic concepts of optimization of real-world systems
- Formulation of a problem as an optimization problem
- Optimality conditions of optimum design with application to simple problems
- Optimization of linear systems: Linear programming
- Optimization of nonlinear systems: Quadratic programming
- Solution of optimization problems using programs

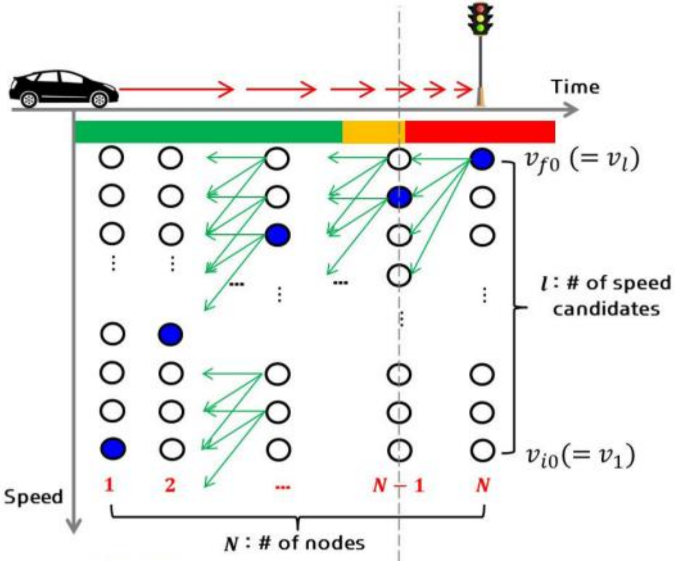
CONTENTS

1. Introduction to Optimization

Optimization is everywhere in real world…

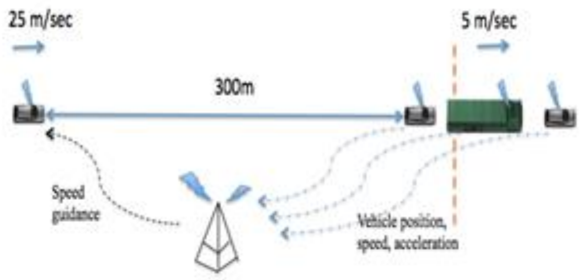
- ❑ Optimal cooperative platoon formation
- ❑ Energy-optimal regenerative braking system





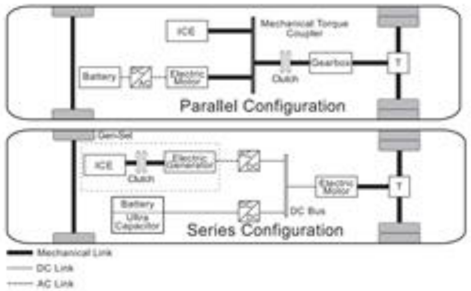
1. Introduction to Optimization

❑ Optimal fleet trajectory control



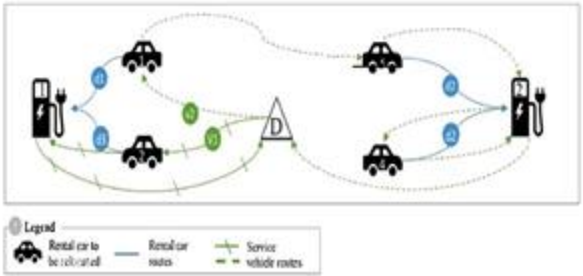
❑ Optimal rebalancing with waiting time

❑ Optimal energy management



❑ Optimal charging (congestion avoidance)

❑ Optimal charging (cost reduction)



❑ Optimal shipment policy

1. Introduction to Optimization

So, what exactly is Optimization?

Optimization is derived from the Latin word “optimus”, **the best**.

Optimization characterizes the activities involved **to find “the best”.**

Definition: Optimization is the mathematical discipline which is concerned with finding the minima and maxima of functions, where the function represents the effort required or the desired benefit. It is an act of obtaining the best results under given circumstances.

1. Introduction to Optimization

- Crucial components in optimization

Objective function: The quantity (or quantities) that one is trying to optimize. It is sometimes referred to as a target.

Optimization variables: These are the variables one can change (sometimes called the changing variables) in order to achieve the optimum solution.

Maximize (Minimize): In some optimization problems, one seeks to make the objective function as large (small) as possible. Such problems are maximization (minimization) problems.

Constraints: Some restrictions among system during the optimization process.

1. Introduction to Optimization

➢ Optimality criteria

In considering optimization problems, two questions generally must be addressed:

- **Static question:** How can one determine whether a given point x^* is the optimal solution?
- **Dynamic question:** If x^* is not the optimal point, then how does one go about finding a solution that is optimal?

1. Introduction to Optimization

➢ General ideas of optimization

There are two ways of examining optimization.

- **Maximization (e.g., maximize profit)**

  In this case one is looking for the highest point on the objective function.

- **Minimization (e.g., minimize cost)**

  In this case one is looking for the lowest point on the objective function.

1. Introduction to Optimization

- Classification of optimization problems

  - Classification based on the existence of the constraints

    unconstrained;   constrained

  - Classification based on physical structure of the problem

    optimal control; non optimal control

  - Classification based on the nature of the equations involved

    linear; nonlinear

  - Classification based on permissible values of the design variables

    integer; real valued programming problems; mixed

1. Introduction to Optimization

- Classification of optimization problems

  - Classification based on no of objective functions

    single; multi objective programming problems
  - Classification based on permissible values of the design variable

    deterministic; stochastic programming
  - Classification based on capability of the search algorithm

    local minimum (maximum); minimum (maximum)
  - Classification based on type of solution.

    analytical/graphical/experimental/numerical methods; search methods

1. Introduction to Optimization

- Primary application areas

Optimization theory finds ready application in all branches of engineering in:

- Design of components or entire systems
- Planning and analysis of existing operations
- Engineering analysis and data reduction
- Control of dynamic systems
- …

1. Introduction to Optimization

➢ Goals

We use optimization to obtain:

- Minimal cost
- Maximum profit
- Best approximation
- Optimal design
- Optimal management or control
- …

CONTENTS

Linear Program: Definition

- A special case of convex optimization problem
- An optimization problem whose optimization objective is a *linear* function and feasible region is defined by a set of *linear* constraints

Variables $x_1, x_2, \ldots, x_n$ that take real values.

LP:

Maximize (or minimize) a *linear objective function*

$$c_1 x_1 + c_2 x_2 + \cdots + c_n x_n$$

Subject to *linear constraints*

$$\begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n &\leq b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n &= b_2 \\ a_{3,1}x_1 + a_{3,2}x_2 + \cdots + a_{3,n}x_n &\geq b_3 \\ &\vdots \end{aligned}$$

$$\max_x c^T x$$
$$\text{s.t. } Gx \leq b$$

$c \in \mathbb{R}^n$
$G \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$

A problem of this form is called a **linear program**.

Example: Nature Connection-Recreational Sites

Nature Connection is planning two new public recreational sites: **a forested wilderness area** and a **sightseeing and hiking park**. They own 80 hectares of forested wilderness area and 20 hectares suitable for the sightseeing and hiking park but they don't have enough resources to make the entire areas available to the public. They have a **budget of $120K** per year. They estimate a yearly management and maintenance cost of $1K per hectare for the forested wilderness area, and $4K per hectare for the sightseeing and hiking park. The expected average number of visiting hours a day per hectare are: 10 for the forest and 20 for the sightseeing and hiking park.

Question: How many hectares should Nature Connection allocate to the public sightseeing and hiking park and to the public forested wilderness area, in order to maximize the amount of recreation, (in average number of visiting hours a day for the total area to be open to the public, for both sites**) given their budget constraint?**

Steps in setting up a LP

1. Determine and label the *decision variables*.
2. Determine the objective and use the decision variables to write an expression for the *objective function*.
3. Determine the constraints - *feasible region*.
    1. Determine the *explicit constraints* and write a functional expression for each of them.
    2. Determine the *implicit constraints (e.g., nonnegativity constraints)*.

Formulation of the problem as a Linear Program

1 Decision Variables

$x1$ – # hectares to allocate to the public forested wilderness area

$x2$ – # hectares to allocate the public sightseeing and hiking park

2 Objective Function

Max $10\ x1 + 20\ x2$

3 Constraints

$x1 \le 80$

$x2 \le 20$

$x1 + 4\ x2 \le 120$

$x1 \ge 0;\ x2 \ge 0$ Non-negativity constraints

Formulation of the problem as a Linear Program

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1 Decision Variables

x1 – # hectares to allocate to the public forested wilderness area

X2 – # hectares to allocate the public sightseeing and hiking park

2 Objective Function

Max 10 x1 + 20 x2

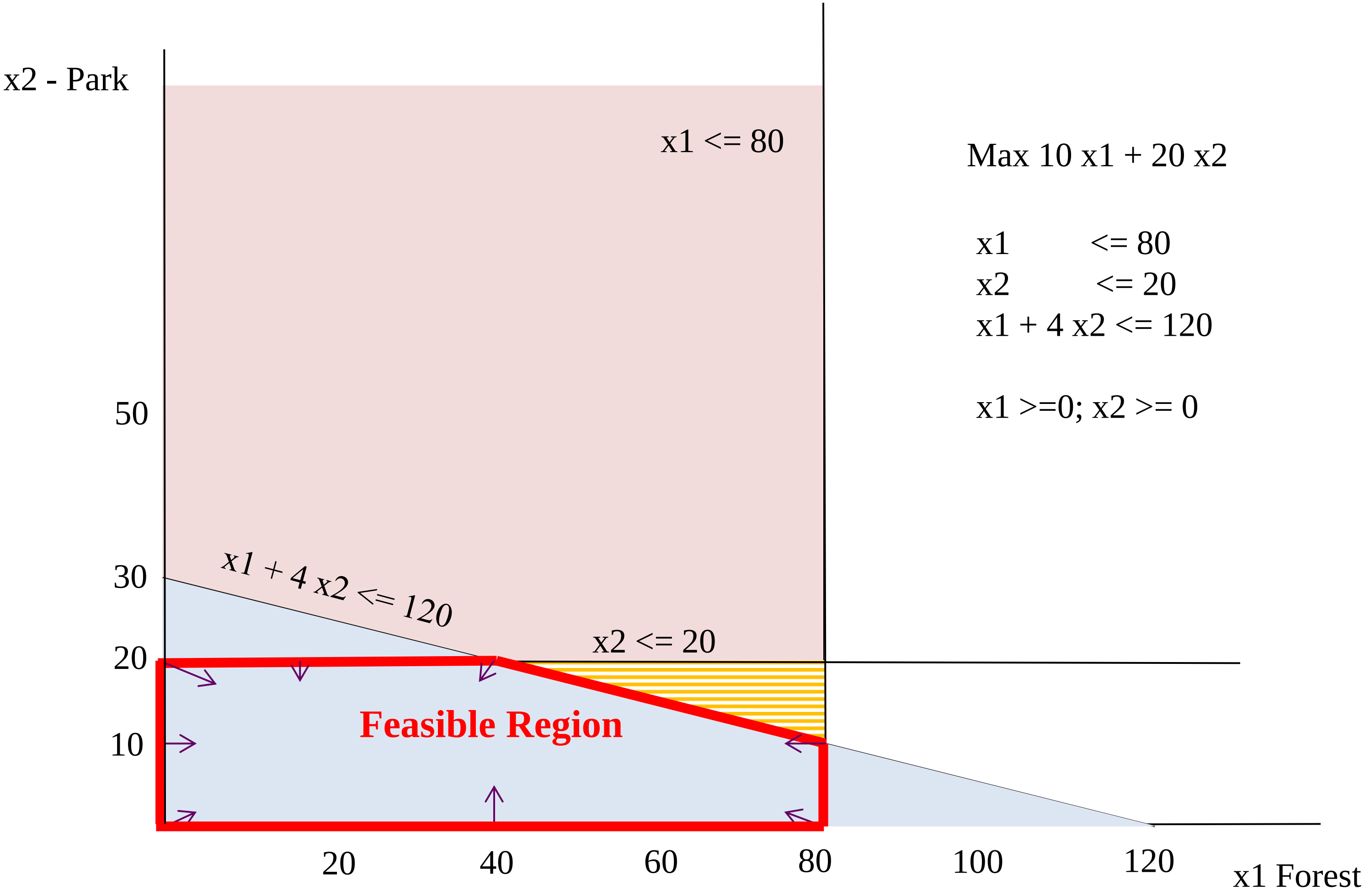
3 Constraints

x1 <= 80 Land for forest

x2 <= 20 Land for Park

x1 + 4 x2 <= 120 Budget

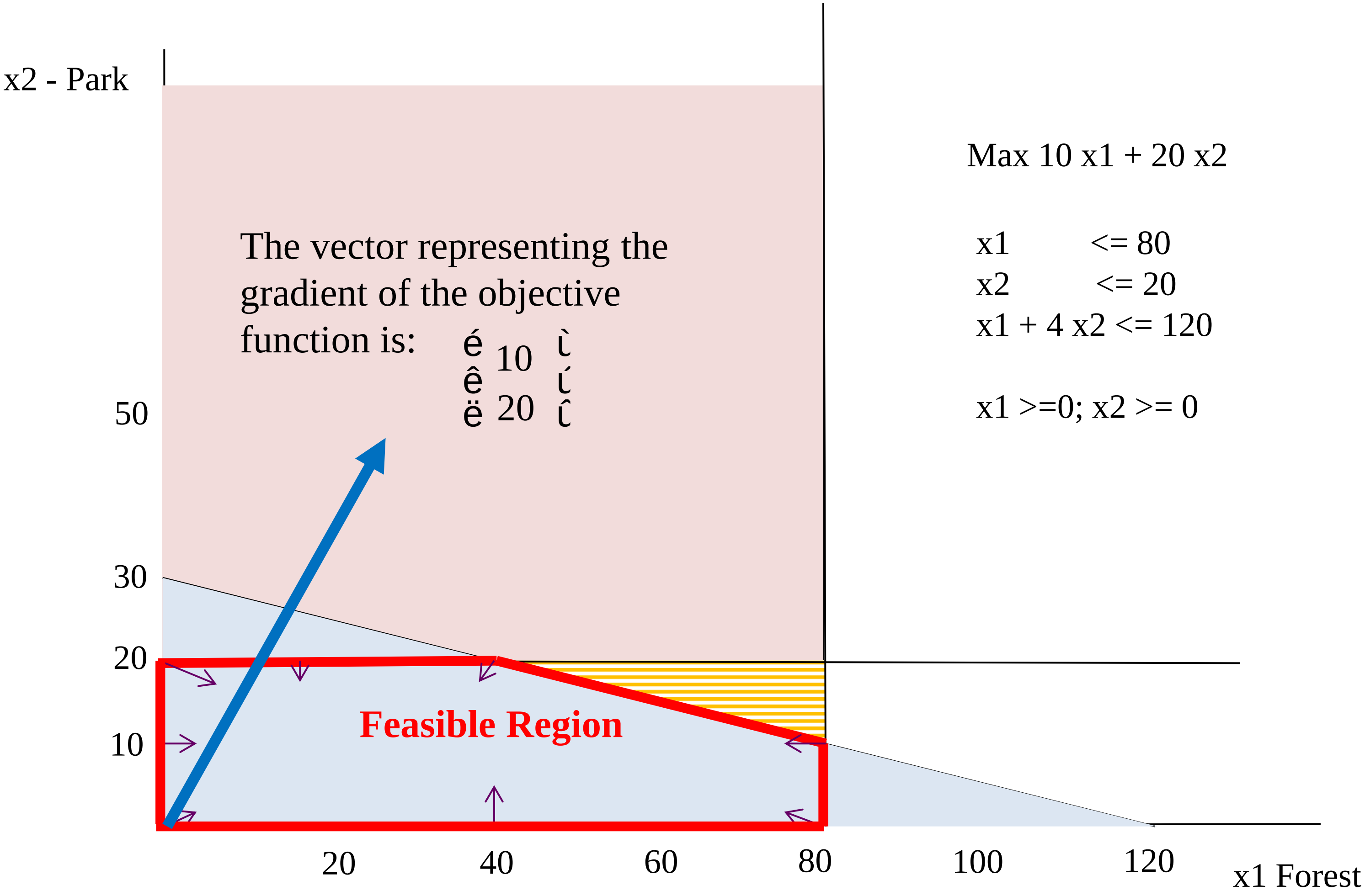
x1 >= 0; x2 >= 0 Non-negativity constraints



Max $10\ x1 + 20\ x2$

$x1 \leq 80$
$x2 \leq 20$
$x1 + 4\ x2 \leq 120$

$x1 \geq 0;\ x2 \geq 0$



Max 10 x1 + 20 x2

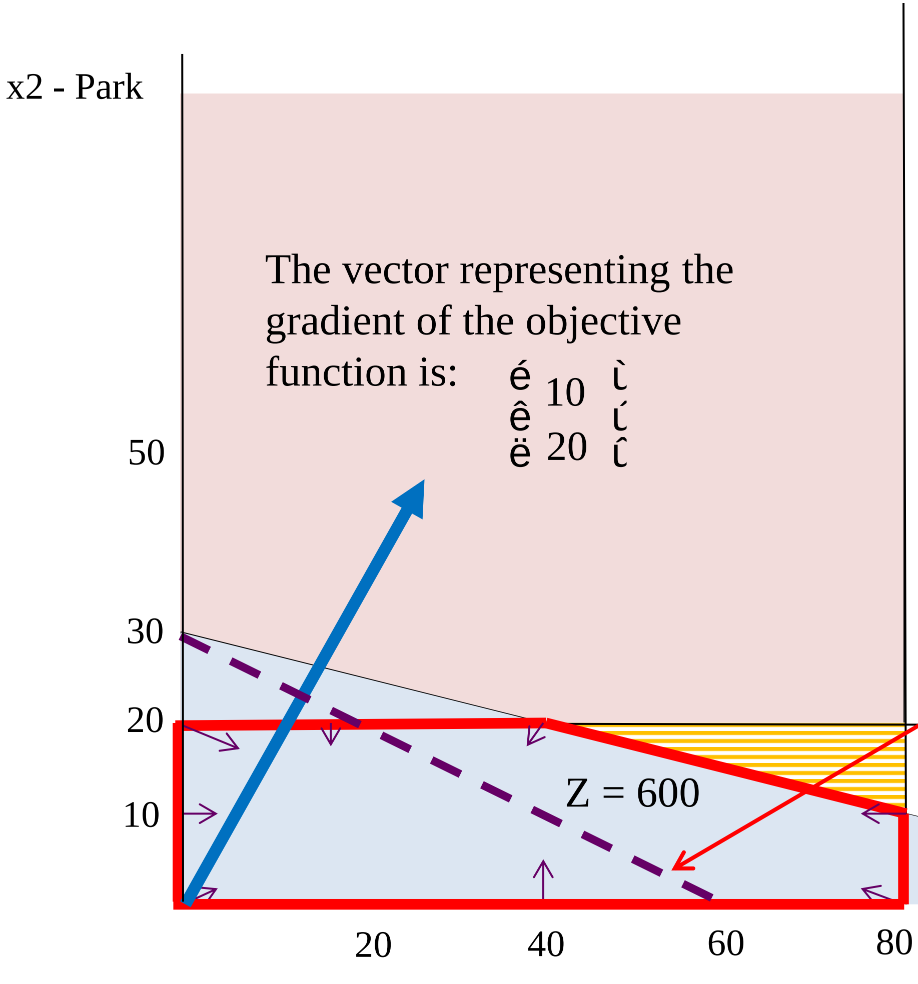
$x1 \le 80$
$x2 \le 20$
$x1 + 4\,x2 \le 120$

$x1 \ge 0;\ x2 \ge 0$

Max 10 x1 + 20 x2

x1 <= 80
x2 <= 20
x1 + 4 x2 <= 120

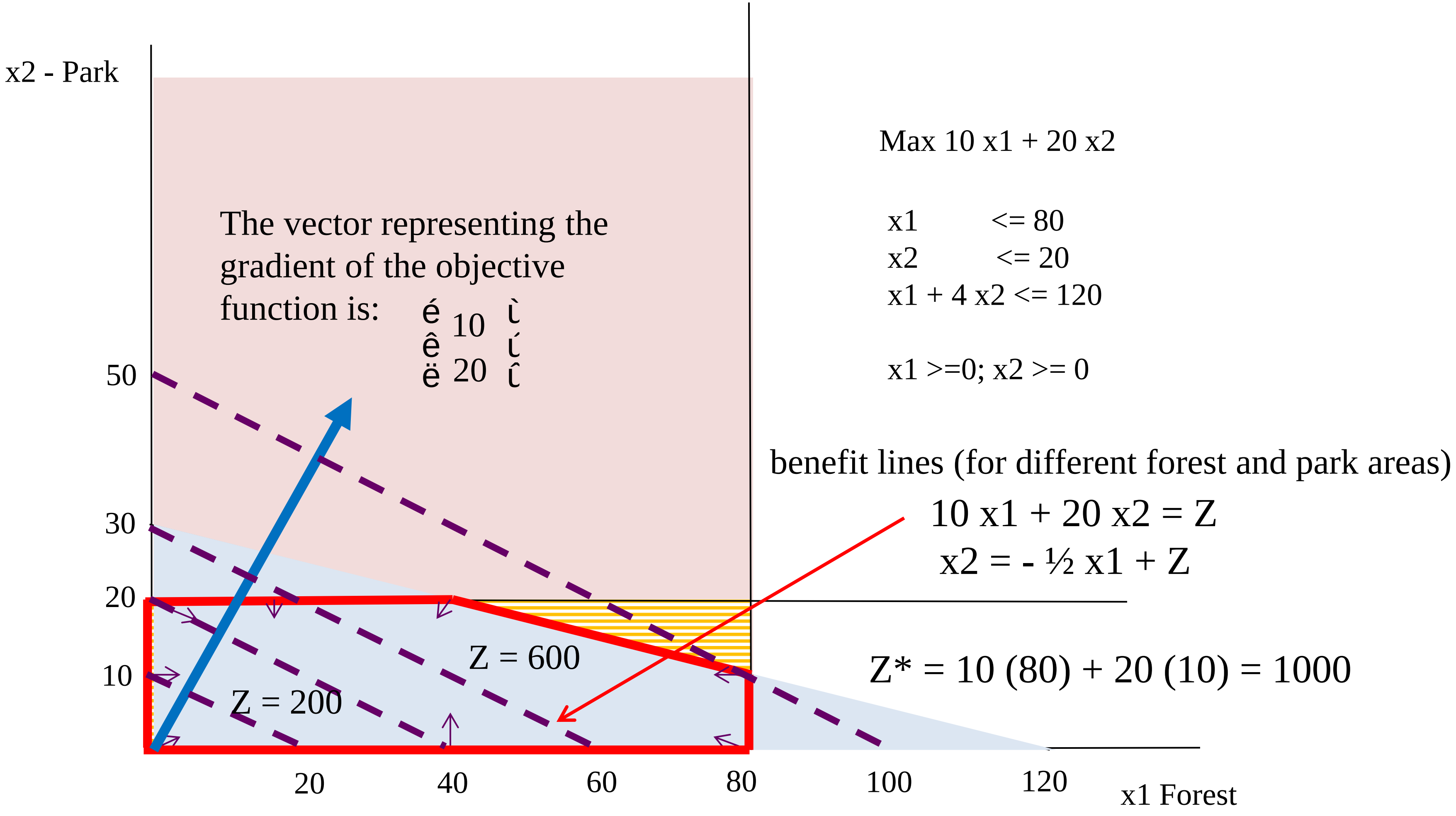
x1 >=0; x2 >= 0



benefit lines (for different forest and park areas)

10 x1 + 20 x2 = Z

x2 = - ½ x1 + Z


x2 - Park
Max 10 x1 + 20 x2
x1 <= 80
x2 <= 20
x1 + 4 x2 <= 120
x1 >=0; x2 >= 0
The vector representing the gradient of the objective function is: $\begin{bmatrix}10\\20\end{bmatrix}$
50
30
20
10
benefit lines (for different forest and park areas)
10 x1 + 20 x2 = Z
x2 = - ½ x1 + Z
Z = 600
Z = 200
Z* = 10 (80) + 20 (10) = 1000
20
40
60
80
100
120
x1 Forest

Summary of the Graphical Method

- Draw the constraint boundary line for each constraint. Use the origin (or any point not on the line) to determine which side of the line is permitted by the constraint.

- Find the feasible region by determining where all constraints are satisfied simultaneously.

- Determine the slope of one objective function line (perpendicular to its gradient vector). All other objective function lines will have the same slope.

- Move a straight edge with this slope through the feasible region in the direction of improving values of the objective function (direction of the gradient). Stop at the last instant that the straight edge still passes through a point in the feasible region. This line is the optimal objective function line.

- A feasible point on the optimal objective function line is an optimal solution.

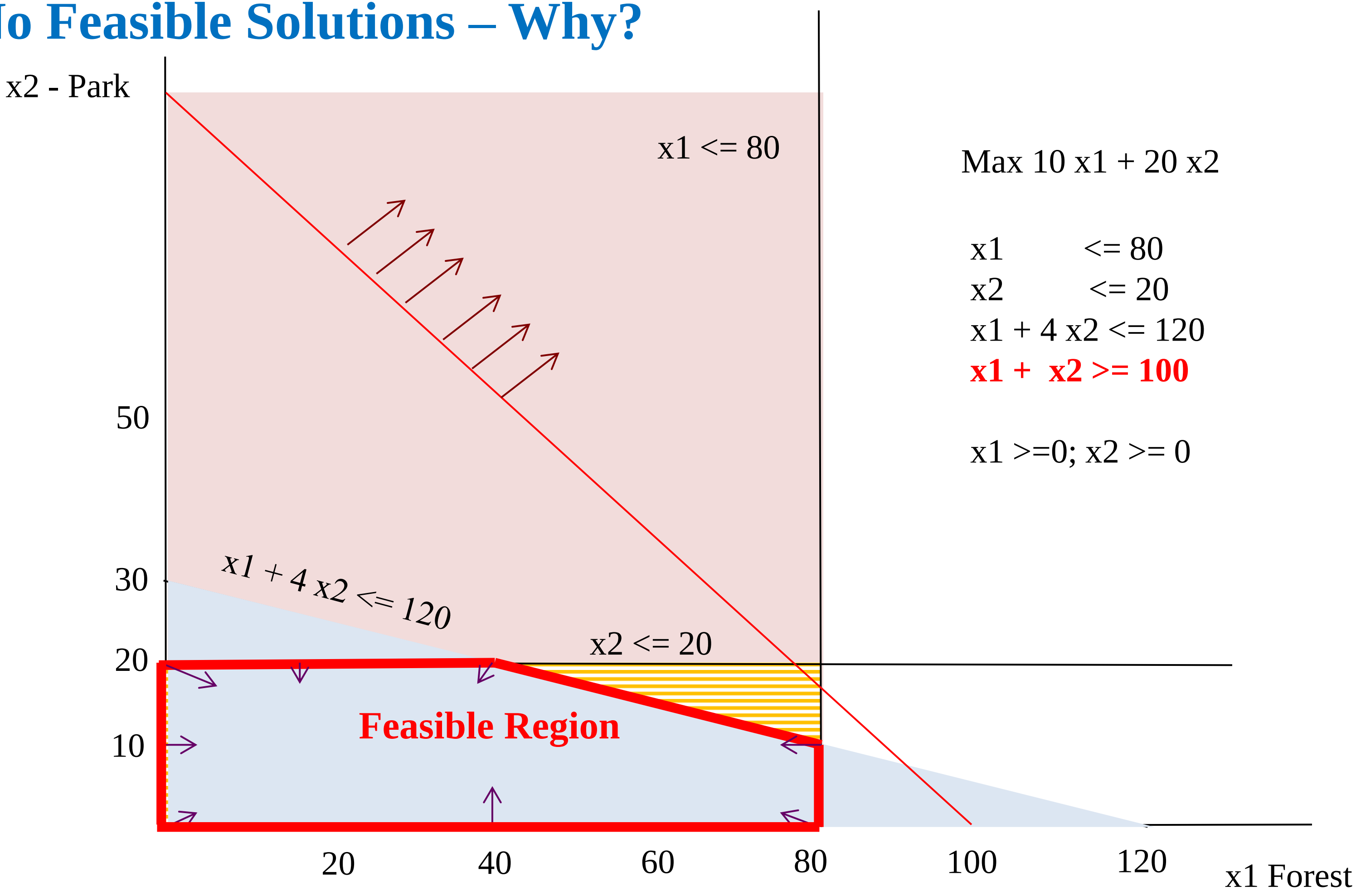
Linear Program: Definition

- Feasible region defined by $Gx \leq b$
  - $g_i^T x \leq b_i$ define a halfspace
  - Feasible region is the intersection of halfspaces, i.e., a linear polytope

- Objective function $c^T x$
  - Represent a direction of optimization
  - Generally, optimum always occur on a corner, i.e., a vertex of the feasible region

$$\max_x c^T x$$
$$\text{s.t. } Gx \leq b$$

No Feasible Solutions – Why?

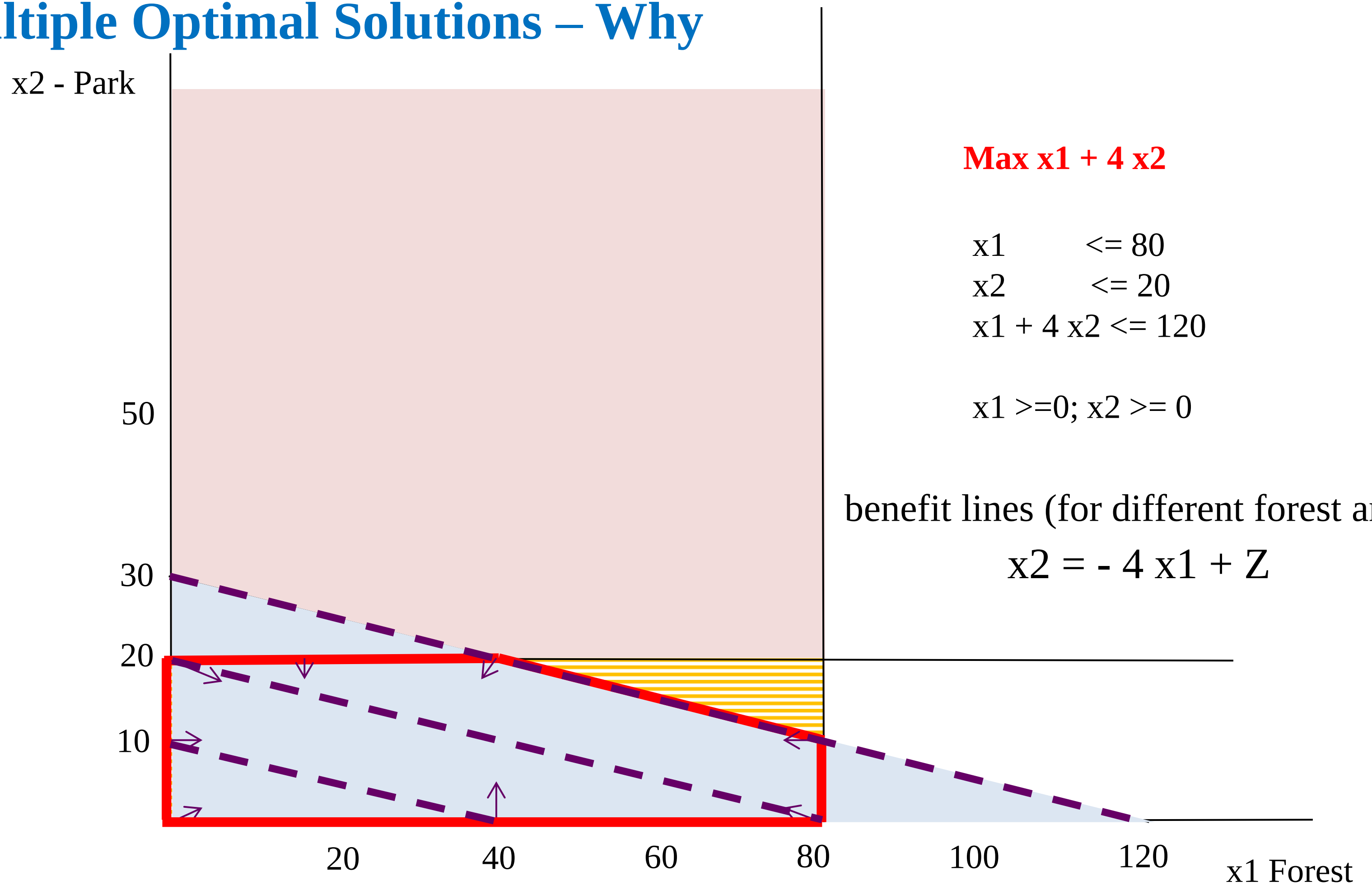


Max 10 x1 + 20 x2

x1 <= 80
x2 <= 20
x1 + 4 x2 <= 120
x1 + x2 >= 100

x1 >=0; x2 >= 0

Multiple Optimal Solutions – Why



Max x1 + 4 x2

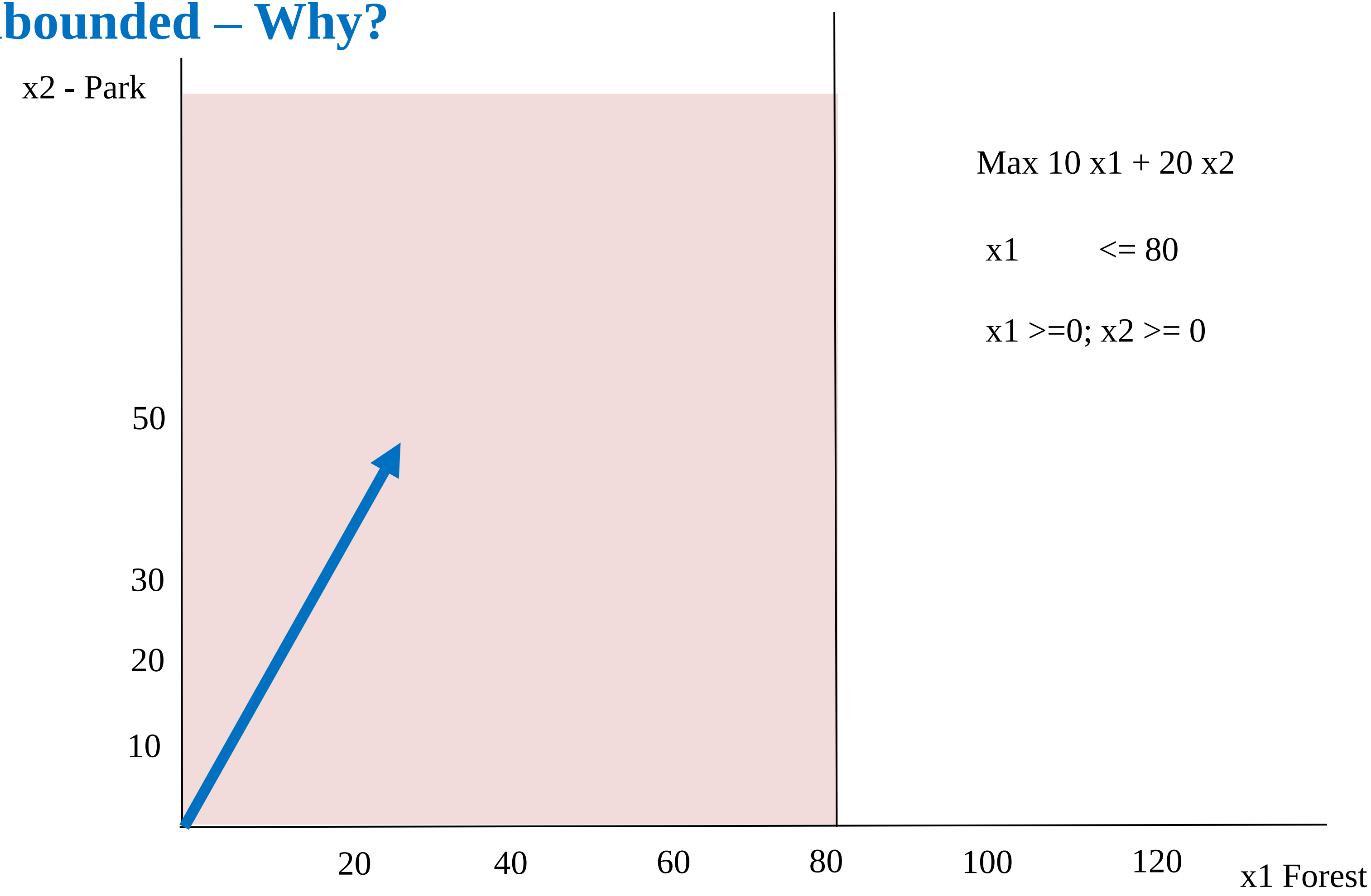
x1 <= 80
x2 <= 20
x1 + 4 x2 <= 120

x1 >=0; x2 >= 0

benefit lines (for different forest and park areas)

x2 = - 4 x1 + Z

Unbounded – Why?



Max 10 x1 + 20 x2

x1 <= 80

x1 >=0; x2 >= 0

Example: Keeping the river clean

Cost-benefit-trade-off problems

Choose the mix of levels of various activities to achieve minimum acceptable levels for various benefits at a minimum cost.

Example: Keeping the river clean

A pulp mill in Maine makes mechanical and chemical pulp, polluting the river in which it spills its spent waters. This has created several problems, leading to a change in management.

The previous owners felt that it would be too expensive to reduce pollution, so they decided to sell the pulp mill. The mill has been bought back by the employees and local businesses, who now own the mill as a cooperative. The new owners have several objectives:

1 – to keep **at least 300 people employed at the mill** (300 workers a day);

2 – to generate at least $40,000 of revenue a day

They estimate that this will be enough to pay operating expenses and yield a return that will keep the mill competitive in the long run. Within these limits, everything possible should be done to **minimize pollution**.

Both chemical and mechanical pulp require the labor of one worker for 1 day (1 workday, wd) per ton produced;

Mechanical pulp sells at $100 per ton; Chemical pulp sells at $200 per ton;

Pollution is measured by the biological oxygen demand (BOD). One ton of mechanical pulp produces 1 unit of BOD; One ton of chemical pulp produces 1.5 units of BOD.

The maximum capacity of the mill to make mechanical pulp is 300 tons per day; for chemical pulp is 200 tons per day. The two manufacturing processes are independent (I.e., the mechanical pulp line cannot be used to make chemical pulp and vice versa).

- Pollution, employment, and revenues result from the production of both types of pulp. So a natural choice for the variables is:

Decision Variables

- X1 amount of mechanical pulp produced (in tons per day, or t/d) and
- X2 amount of chemical pulp produce (in tons per day, or t/d

Min Z =	1 X1	+	1.5 X2	
(BOD/day)	(BOD/t) (t/d)		(BOD/t) (t/d)	
Subject to:				
	1 X1	+	1 X2	>= 300 Workers/day
	(wd/t) (t/d)		(wd/t)(t/d)	
	100 X1	+	200 X2	>= 40,000 revenue/day
	($/t) (t/d)	+	($/t)(t/d)	$/d
	X1			<= 300 (mechanical pulp)
	(t/d)			(t/d)
			X2	<= 200 (mechanical pulp)
			(t/d)	(t/d)
	X1 >=0; X2 >= 0			

Example: Distribution-Network Problem

- The International Hospital Share Organization is a non-profit organization that refurbishes a variety of used equipment for hospitals of developing countries at two international factories (F1 and F2). One of its products is a large X-Ray machine.

- Orders have been received from three large communities for the X-Ray machines.

Example: Distribution-Network Problem

	Shipping Cost for Each Machine (Model L)			
To	Community 1	Community 2	Community 3	
From				**Output**
Factory 1	$700	$900	$800	12 X-ray machines
Factory 2	800	900	700	15 X-Ray machines
Order Size	10 X-ray machines	8 X-Ray machines	9 X-Ray machines	

Question: How many X-Ray machines (model L) should be shipped from each factory to each hospital so that shipping costs are minimized?

Example: Distribution-Network Problem

- Activities – shipping lanes (not the level of production which has already been defined)
  - Level of each activity – number of machines of model L shipped through the corresponding shipping lane.
  - Best mix of shipping amounts

Requirement 1: Factory 1 must ship 12 machines
Requirement 2: Factory 2 must ship 15 machines
Requirement 3: Customer 1 must receive 10 machines
Requirement 4: Customer 2 must receive 8 machines
Requirement 5: Customer 3 must receive 9 machines

Example: Distribution-Network Problem

Let S_{ij} = Number of machines to ship from i to j (i = F1, F2; j = C1, C2, C3).

$$\text{Minimize Cost} = \$700S_{F1\text{-}C1} + \$900S_{F1\text{-}C2} + \$800S_{F1\text{-}C3} + \$800S_{F2\text{-}C1} + \$900S_{F2\text{-}C2} + \$700S_{F2\text{-}C3}$$

subject to

Factory 1: $S_{F1\text{-}C1} + S_{F1\text{-}C2} + S_{F1\text{-}C3} = 12$

Factory 2: $S_{F2\text{-}C1} + S_{F2\text{-}C2} + S_{F2\text{-}C3} = 15$

Customer 1: $S_{F1\text{-}C1} + S_{F2\text{-}C1} = 10$

Customer 2: $S_{F1\text{-}C2} + S_{F2\text{-}C2} = 8$

Customer 3: $S_{F1\text{-}C3} + S_{F2\text{-}C3} = 9$

and

$S_{ij} \geq 0$ (i = F1, F2; j = C1, C2, C3).

Linear Program: How to Solve

- Naïve approach
  - Find all vertices
  - Pick the one with the best objective value
- Practical approach: Simplex algorithm
  - Start from one vertex
  - Move towards a neighboring vertex for improvement
  - Recall local search and gradient descent
  - May enumerate almost all the vertices in the worst case, but very efficient in most cases
- (Not covered) Ellipsoid algorithm
  - Polynomial-time in theory, poor performance in practice

Linear Program: Standard Form

- To solve LPs, typically need to put them in standard form:

$$\begin{aligned} &\max\ \boldsymbol{c}^T\boldsymbol{x} \\ \text{s.t.}\ \ &A\boldsymbol{x} \leq \boldsymbol{b} \\ &\boldsymbol{x} \geq 0 \end{aligned}$$

All constraints are $\leq$ inequalities.

All variables are constrained to be *non-negative*.
(Very special linear inequalities.)

- Every LP can be easily converted to standard form.

$$\min \sum_{j=1}^{n} c_j x_j \quad \Leftrightarrow \quad \max \sum_{j=1}^{n} (-c_j) x_j$$

$$\sum_{j=1}^{n} a_{i,j} x_j \geq b_i \quad \Leftrightarrow \quad \sum_{j=1}^{n} (-a_{i,j}) x_j \leq -b_i$$

$$x \lesseqgtr 0 \quad \Leftrightarrow \quad \begin{aligned} &x = x^+ - x^- \\ &x^+, x^- \geq 0 \end{aligned}$$

Linear Program: How to Solve in Practice

Solve LPs in practice:

- All the solvers/algorithms for Convex Optimization problems can be applied
- Additional solvers/algorithms for LPs
  - linprog (MATLAB), linprog (in Python package SciPy)
  - PuLP (Python)
  - Cplex, Gurobi
  - https://www.informs.org/ORMS-Today/Public-Articles/June-Volume-38-Number-3/Software-Survey-Linear-Programming

Matlab code for an example

```
%       OctRt          Price $/B
Gas = [93     37.5;
          85     28.5];
%Stock        OctRt              Price $/B     Availability
Stock =   [70                     12.5          2000;
              80                  12.5          4000;
              85                  12.5          4000;
              90                  27.5          5000;
              99                  27.5          3000];
% Revenue
f = [zeros(10,1);     Stock(:,3);   Gas(:,2)];
% Equality constraint
G = [eye(5,5)        eye(5,5)      eye(5,5)      zeros(5,2);
        ones(1,5)     zeros(1,5)    zeros(1,5)    -1      0;
        zeros(1,5)    ones(1,5)     zeros(1,5)    0       -1];
h = [Stock(:,3);      zeros(2,1)];
% Inequality (fuel quality) constraints
A = [-[Stock(:,1)'            zeros(1,5)            zeros(1,5);
        zeros(1,5)            Stock(:,1)'           zeros(1,5)]  diag(Gas(:,1))];
b = zeros(2,1);
```

```
% X=LINPROG(f,A,b,Aeq,beq,LB,UB)
x = linprog(-f,A,b,G,h,zeros(size(f)),[]);
Revenue = f'*x
```

Linear Program: Additional Resources

- Textbook

  Applied Mathematical Programming, Chapters 2-4

  By Bradley, Hax, and Magnanti (Addison-Wesley, 1977)

  http://web.mit.edu/15.053/www/AMP.htm

- Online course

  https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-251j-introduction-to-mathematical-programming-fall-2009/index.htm

Summary:

- What is LP.
- How to formulate LP
- The graphical method
- How to solve LP

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THANKS